

DHRUV CHOUDHARY

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SUMMARY

AI/ML Engineer specializing in **Generative AI, Agentic AI, and machine learning**, with hands-on experience building **AI applications, Large Language Models (LLMs), RAG systems, computer vision systems, recommendation engines, REST APIs, and ML inference services**. Proficient in **Python, FastAPI, LLM integration, RAG, React, ChromaDB, MCP, LangGraph, and model deployment**, with experience developing end-to-end AI systems from backend architecture to user-facing interfaces.

EDUCATION

VIT Bhopal University

Bachelor of Technology in Computer Science and Engineering – AI & ML

Bhopal, India

Sep. 2023 – Current

PROJECTS

Agentic Email Assistant | Python, LLMs, MCP, LangGraph, React, Vite, Tailwind

2026

- Built an autonomous **AI email-triage agent in Python** using an agent loop for inbox analysis, structured planning, tool execution, state management, logging, and human-in-the-loop approval for safe automation.
- Engineered **Large Language Model (LLM)-driven planning and tool orchestration** with constrained JSON outputs, prompt engineering, tool executors, and MCP client wrappers for controlled email actions and persistent memory management.
- Developed a responsive **React + Vite + Tailwind frontend** with Axios-based API integration and backend unit tests, enabling end-to-end development, testing, and local deployment.

HARMONY – AI Health Assistant | FastAPI, Python, RAG, ChromaDB, Telegram API

2025

- Built a **FastAPI-based AI health assistant** with a multilingual RAG pipeline using ChromaDB and sentence-transformers, supporting English, Hindi, Spanish, French, and Arabic interactions.
- Implemented **machine learning-based disease prediction and healthcare information workflows**, including symptom-based prediction and nearby hospital discovery using live OpenStreetMap/Nominatim data.
- Integrated a **Telegram bot and responsive web interface** supporting symptom queries, medicine information, voice interactions, medical-report workflows, and basic health logging.

Smart Retail AI Platform | FastAPI, Python, OpenCV, TensorFlow, Keras, NumPy, Joblib

2025

- Built a modular **FastAPI REST backend** integrating **computer vision systems**, recommendation, forecasting, segmentation, and fraud-detection services through production-ready inference endpoints.
- Developed **computer vision and recommendation pipelines** using OpenCV, TensorFlow, Keras, NumPy, and Joblib, implementing product classification, image preprocessing, collaborative filtering, efficient top-*N* scoring, and popularity-based cold-start recommendations.
- Deployed **machine learning inference services** for forecasting and fraud detection using serialized `.pkl` and `.keras` models, enabling end-to-end model serving through REST APIs with robust error handling.

TECHNICAL SKILLS

- **Programming:** Python, Java, JavaScript (ES6+), C/C++, SQL, HTML/CSS
- **AI / Machine Learning:** Machine Learning, Large Language Models (LLMs), Generative AI, Agentic AI, RAG, Prompt Engineering, Computer Vision, Vision Systems, Recommendation Systems, Forecasting, Fraud Detection, NLP
- **Frameworks & Libraries:** FastAPI, Flask, React, Node.js, Express.js, OpenCV, TensorFlow, Keras, NumPy, pandas, Matplotlib, sentence-transformers
- **AI / Data Technologies:** LangGraph, ChromaDB, MCP, Joblib, REST APIs, Telegram Bot API, OpenAI API, OpenStreetMap/Nominatim
- **Developer Tools:** Git, GitHub, Docker, VS Code, PyCharm, IntelliJ IDEA
- **Cloud / Platforms:** Google Cloud Platform, AWS

CERTIFICATIONS

Machine Learning Specialization

Coursera

Stanford University & DeepLearning.AI

- Credential: coursera.org/verify/specialization/OPB31TQG7BB2